

Automated Network Scanner & Web Dashboard

Cyber Security Project — Network Reconnaissance Automation

Author:	Karthik Ediga	Target Subnet:	10.0.2.0/24
Environment:	Kali Linux (Apache2, Cron)	Core Tools:	Netdiscover, Masscan, Nmap

1. Environment Setup & Tool Installation

```
sudo apt update && sudo apt install nmap netdiscover masscan apache2 -y
sudo systemctl start apache2 && sudo systemctl enable apache2
```

2. Bash Scanner Script (/home/kali/network_scanner.sh)

```
#!/bin/bash

# Define output location and current timestamp
OUTPUT_FILE="/var/www/html/index.html"
TIMESTAMP=$(date +%Y-%m-%d %H:%M:%S)

# 1. ARP Host Discovery with Netdiscover
NETDISCOVER_OUT=$(sudo netdiscover -i eth0 -r 10.0.2.0/24 -P 2>&1)

# 2. Fast Port Sweep with Masscan
MASSCAN_OUT=$(sudo masscan 10.0.2.0/24 -p80,443,22,135,3306 --router-ip 10.0.2.2 --rate=1000 2>&1 | grep Discovered)

# 3. Detailed Service Auditing with Nmap
NMAP_OUT=$(nmap -F 10.0.2.0/24)

# Generate the dynamic HTML dashboard
cat <<EOF > $OUTPUT_FILE
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>Automated Network Port Scan Results</title>
</head>
<body>
  <h1[!] Automated Network Port Scan Results</h1>
  <p><b>System Time:</b> $TIMESTAMP</p>
  <h3>Target Scan Scope: 10.0.2.0/24</h3>

  <h2>1. Active Host Discovery (Netdiscover)</h2>
  <pre>
$NETDISCOVER_OUT
  </pre>

  <h2>2. High-Speed Port Sweep (Masscan)</h2>
  <pre>
$MASSCAN_OUT
  </pre>

  <h2>3. Detailed Service Scan (Nmap)</h2>
  <pre>
$NMAP_OUT
```

```
</pre>
```

```
<p>Kali Linux Lab Utility – Refreshes automatically every 10 minutes via Cron.</p>
```

```
</body>
```

```
</html>
```

```
EOF
```

3. Script Permissions & Cron Automation

```
# Grant execution permissions
```

```
chmod +x /home/kali/network_scanner.sh
```

```
# Schedule via Crontab (crontab -e) to run every 10 minutes
```

```
*/10 * * * * /home/kali/network_scanner.sh
```

Dashboard URL: `http://localhost/index.html` (or `http://<target-ip>/index.html` across the local subnet)